

Politecnico di Torino

Quantum Hardware Design and Optimization

Laboratory 1

Quantum Gates Introduction

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12 ottobre 2026

Instructions for laboratories submission

Create a folder with the following name:

`gr<nn>_lab1`

where *n* is the group number.
Inside, create a folder for each Exercise:

```
es01
es02
es03
```

containing the developed scripts and eventual files necessary for their execution.
The day before the laboratory evaluation meeting, submit the folder in the Elaborati section of the Portale della Didattica.

1 Evaluation by hand of the state evolution of a circuit

Let's consider the following circuit:

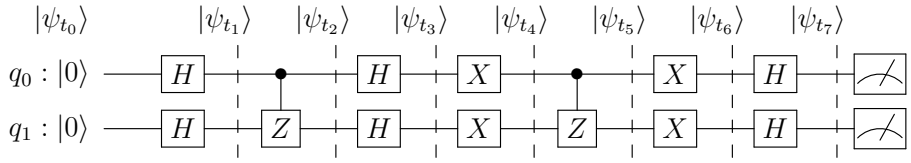


Figura 1: Quantum Circuit.

Compute by hand the state vector evolution layer by layer and try to deduce its aim. The computation of the first layer is reported here as an example.

1.1 Example

The initial state of the system of qubits is computed as:

$$|\psi_{t_0}\rangle = |0\rangle \otimes |0\rangle = \begin{pmatrix}1\\0\end{pmatrix} \otimes \begin{pmatrix}1\\0\end{pmatrix} = \begin{pmatrix}1\\0\\0\\0\end{pmatrix} \rightarrow |00\rangle \ .$$

The equivalent matrix of the first layer is computed as:

$$\mathrm{G}_1 = \frac{1}{\sqrt{2}}\begin{pmatrix}1 & 1\\1 & -1\end{pmatrix} \otimes \frac{1}{\sqrt{2}}\begin{pmatrix}1 & 1\\1 & -1\end{pmatrix} = \frac{1}{2}\begin{pmatrix}1 & 1 & 1 & 1\\1 & -1 & 1 & -1\\1 & 1 & -1 & -1\\1 & -1 & -1 & 1\end{pmatrix} \ .$$

The state $|\psi_{t_1}\rangle$ is evaluated as:

$$|\psi_{t_1}\rangle = \mathrm{G}_1 \, |\psi_{t_0}\rangle = \frac{1}{2}\begin{pmatrix}1 & 1 & 1 & 1\\1 & -1 & 1 & -1\\1 & 1 & -1 & -1\\1 & -1 & -1 & 1\end{pmatrix}\begin{pmatrix}1\\0\\0\\0\end{pmatrix} = \frac{1}{2}\begin{pmatrix}1\\1\\1\\1\end{pmatrix} \rightarrow |{+}{+}\rangle \ .$$

2 Quirk implementation

Now, compose the circuit with Quirk and verify the correctness of your calculus in the first Exercise.

3 Qiskit implementation

Now, implement the circuit with qiskit and simulate it with both state vector and basic simulator.
Finally, generate with qiskit the openqasm description of the circuit and analyze it.

4 Optional

The following exercise is not mandatory.
Repeat the exercises 1, 2, and 3 with the following circuit and try to understand the differences.

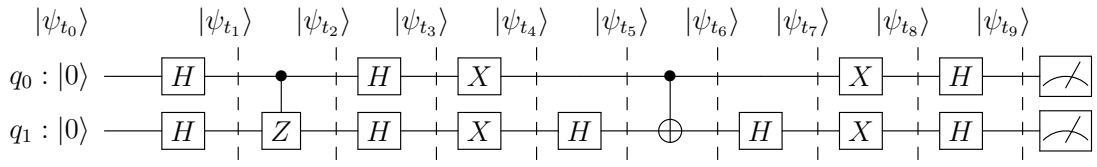


Figura 2: Quantum Circuit optional.

Submission

It is requested to submit:

1. pdf file containing the calculations performed in the exercise 1, named `es01.pdf` with explicit calculus;
2. the offline copy of your circuit, rename as `ex02.html`;
3. the python script describing and simulating the circuit, named `es03.py`, and the openqasm file generated by qiskit, named `es03_qasm.qasm`.
4. for the optional exercises submit the same files as the previous steps but append `_optional` at the end of each file name (e.g. `ex01_optional.pdf`).